



Site Reliability Engineering

Organizational & Operational Framework
(*a.k.a.* “Work Mode”)

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Agenda

Introduction

Weakness / Strengths

Changes

- Culture of Service
- Bi-Modality Awareness
- Ways of Working
 - Organization
 - Innovation
 - Communication
 - Planning

Trust & Visibility

Round Robin (Q&A)



Introduction

Purpose

- Clarify **expectations** from company leaders
- **Align** all the people then address technology issues
- Allow collaborators to **share concerns** (*at the end, take notes*)
- Propose a **set of criteria** used to follow-up teamwork
- Opportunity to be honest, positive and **share ideas**
- Put everyone in the **same (new) page**, no excuses
- **Avoid repeating** this information several times

Site Reliability Engineering

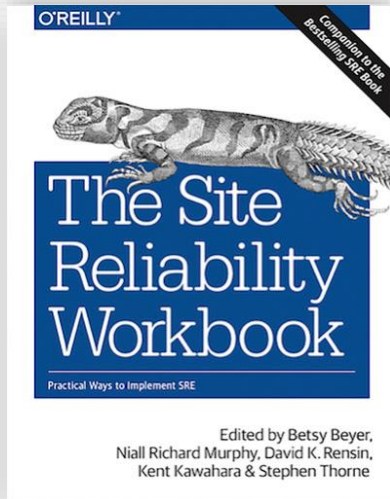
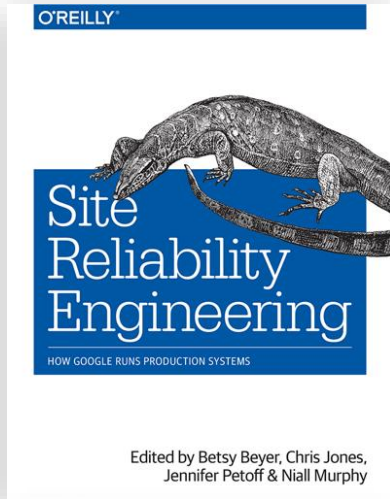
Vision

- *Be recognized as an **expert and reliable group of people** which offers the **best in class knowledge and support** over the core IT infrastructure.*

Mission:

- *Ensure the **continuous and secure operation and support** of the network, compute, storage, backup, messaging, logging and monitoring platforms of the core IT infrastructure enabling the critical business application and services.*

Site Reliability Engineering



Metrics & Monitoring

- SLOs
- Dashboards
- Analytics



Capacity Planning

- Forecasting
- Demand-driven
- Performance



Change Management

- Release process
- Consulting design
- Automation



Emergency Response

- Oncall
- Analysis
- Postmortems



Culture

- Toil management
- Engineering alignment
- Blamelessness

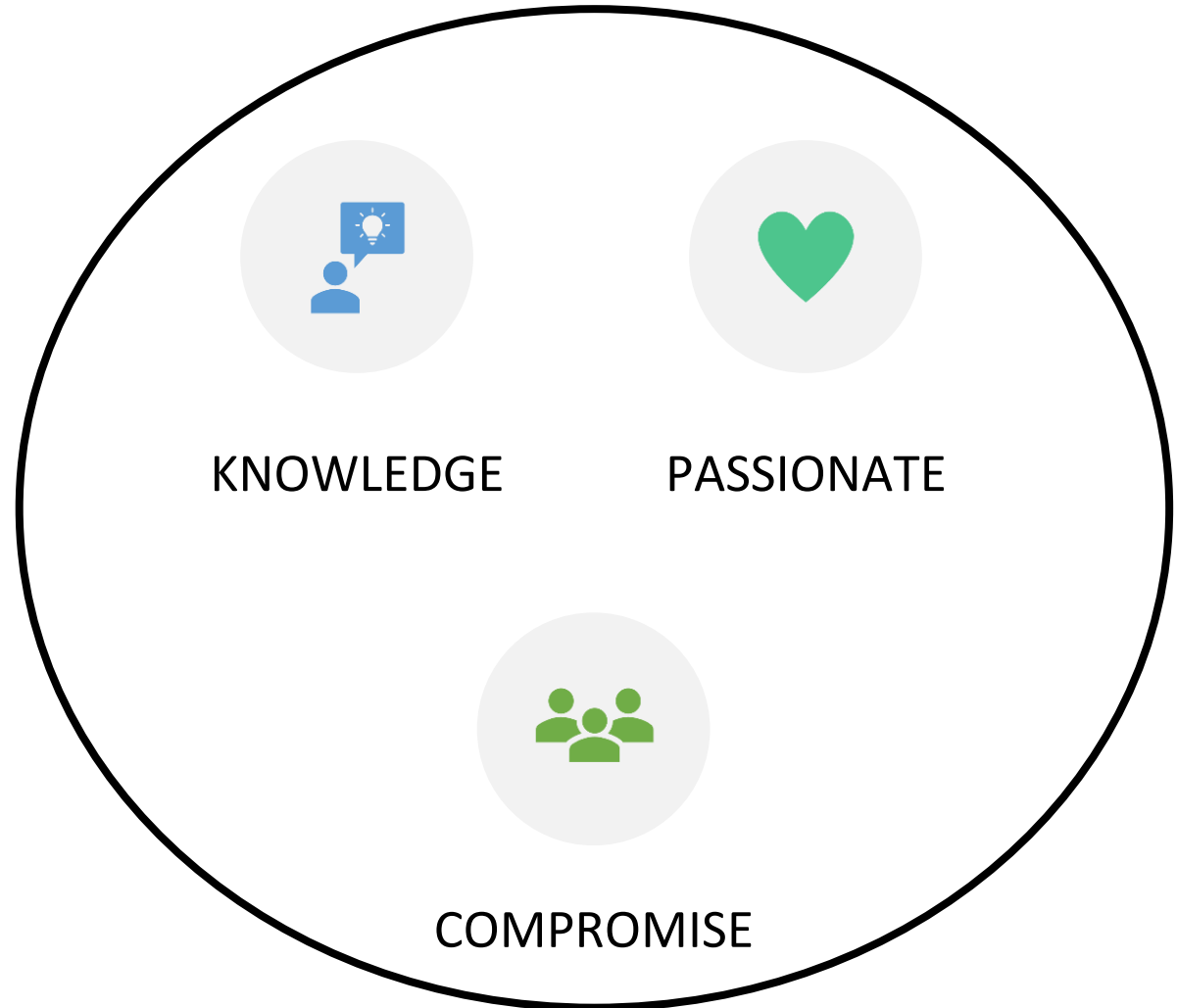


<https://sre.google/books/>



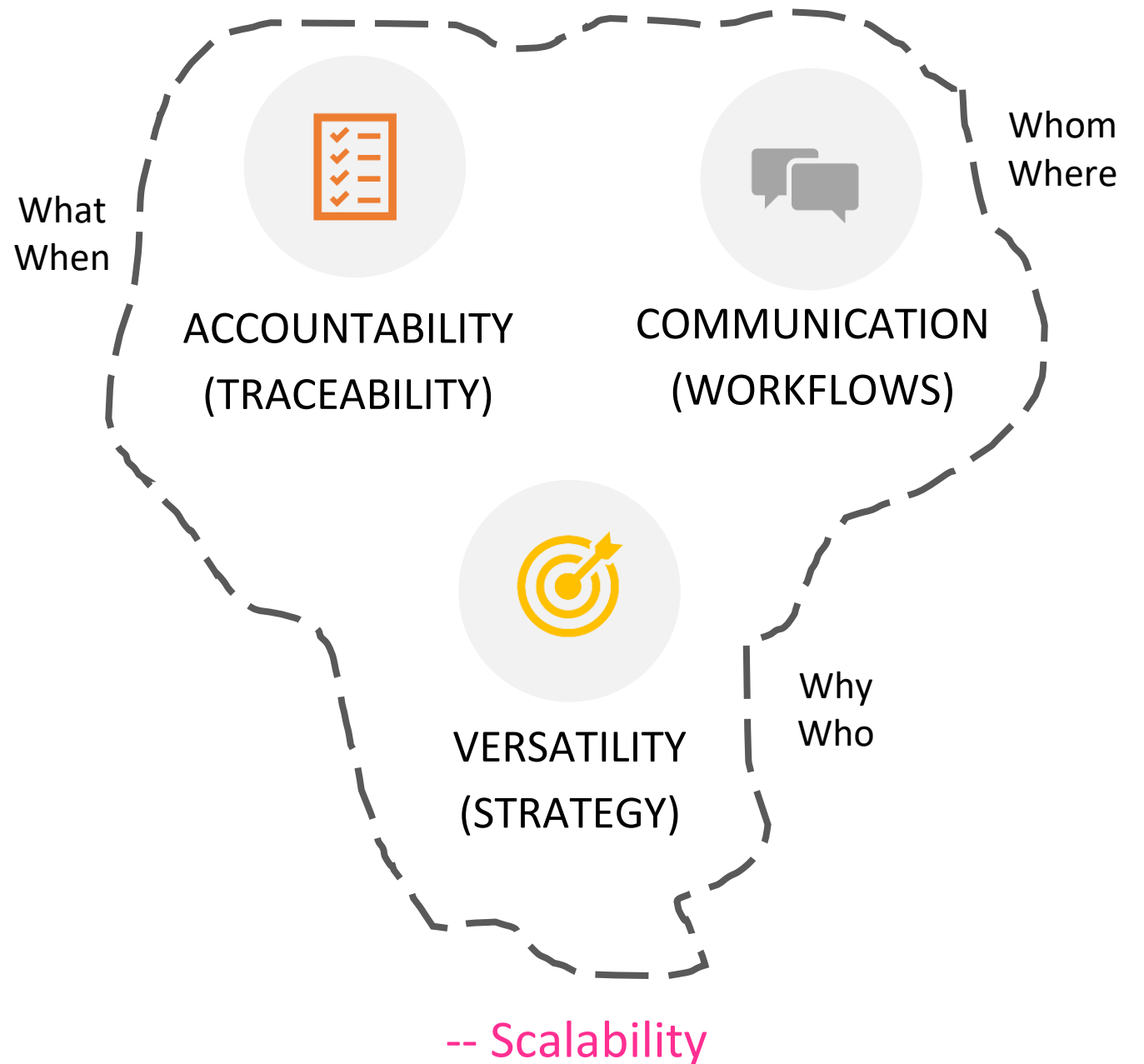
Weakness & Strengths

Strengths



++ Assertiveness
++ Resolution

Weakness





Changes

The Company / Business Culture



gartner.com/SmarterWithGartner

Source: Gartner

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Gartner

Blindness?

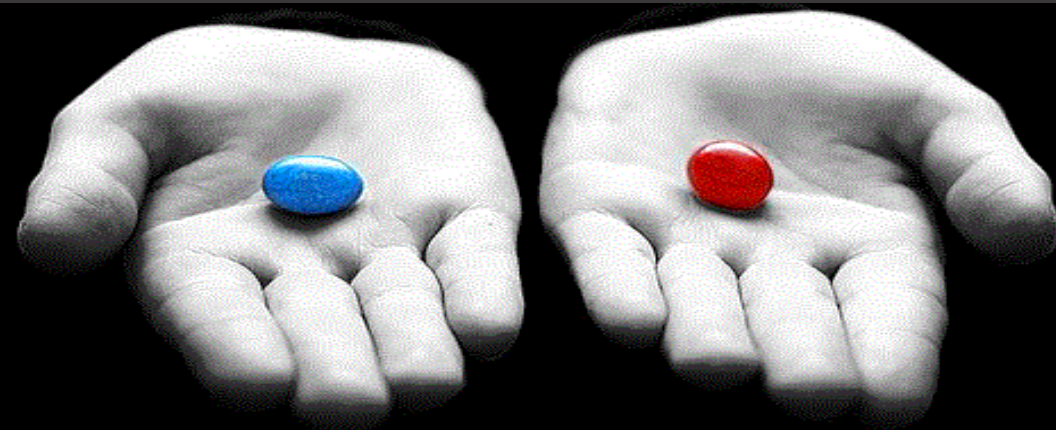


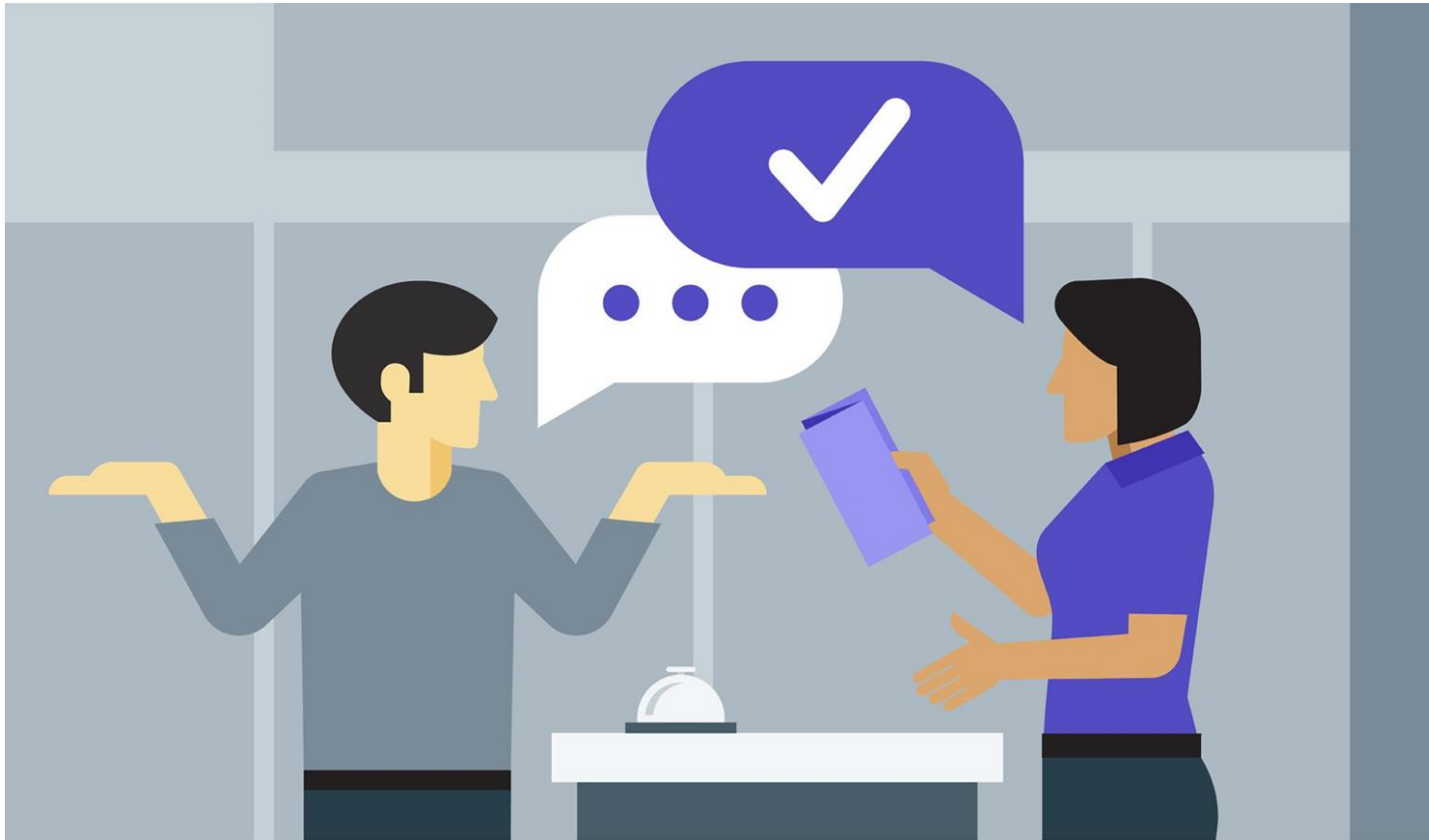
Start with You

**CHANGE
AHEAD**



Vicious Cycle: I really like a different future, but why change if everything is ok the way it is. The (other) people has to change.





- Everybody has a movie
- Criteria (dis-)alignment

Every:

- Business
- Customer
- Organization
- Employee

Has:

1. Opportunities
2. Values
3. Risks
4. ~~Rules~~ Standards

Change

- Incremental
- Iterative
- Agreed



CULTURE
(OF SERVICE)



BI-MODAL
(VERSATILITY)



WAYS OF WORKING
(ME / YOU / ALL)



LEGACY
(HR / IT)



Change
|
Culture
of Service

User Experience => Every Interaction Counts

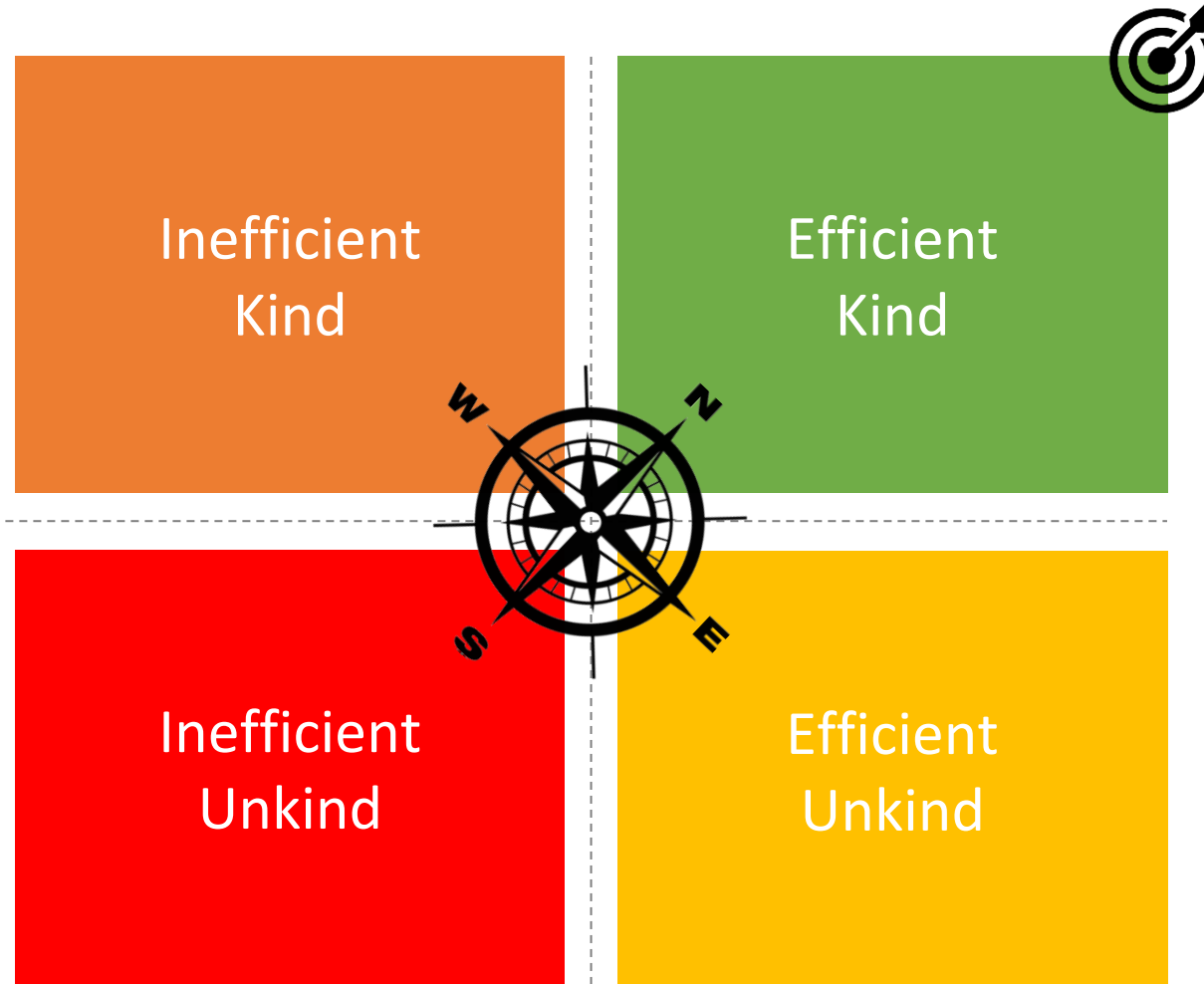


**91% of customers who
had a bad experience
will not do business with
your company again.**

Source: Glance

Customers ⇔ Team / Group / Member / Partner (Internal & External)

Service Delivery Orientation



1. Developers

2. DevOps

3. SysOps

CUSTOMER
FEEDBACK
SUPPORT
INNOVATIVE
QUALITY
EXCELLENT
FRIENDLY



Emotional Intelligence

Conflict $\Leftrightarrow$ Negotiation

Tension $\Leftrightarrow$ Nothing

Conflict $\neq$ Fight



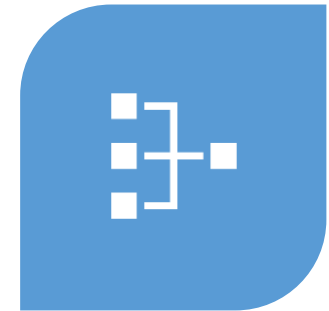
KEY FOR PEOPLE
RELATIONS



KEY FOR WORK
RELATIONS



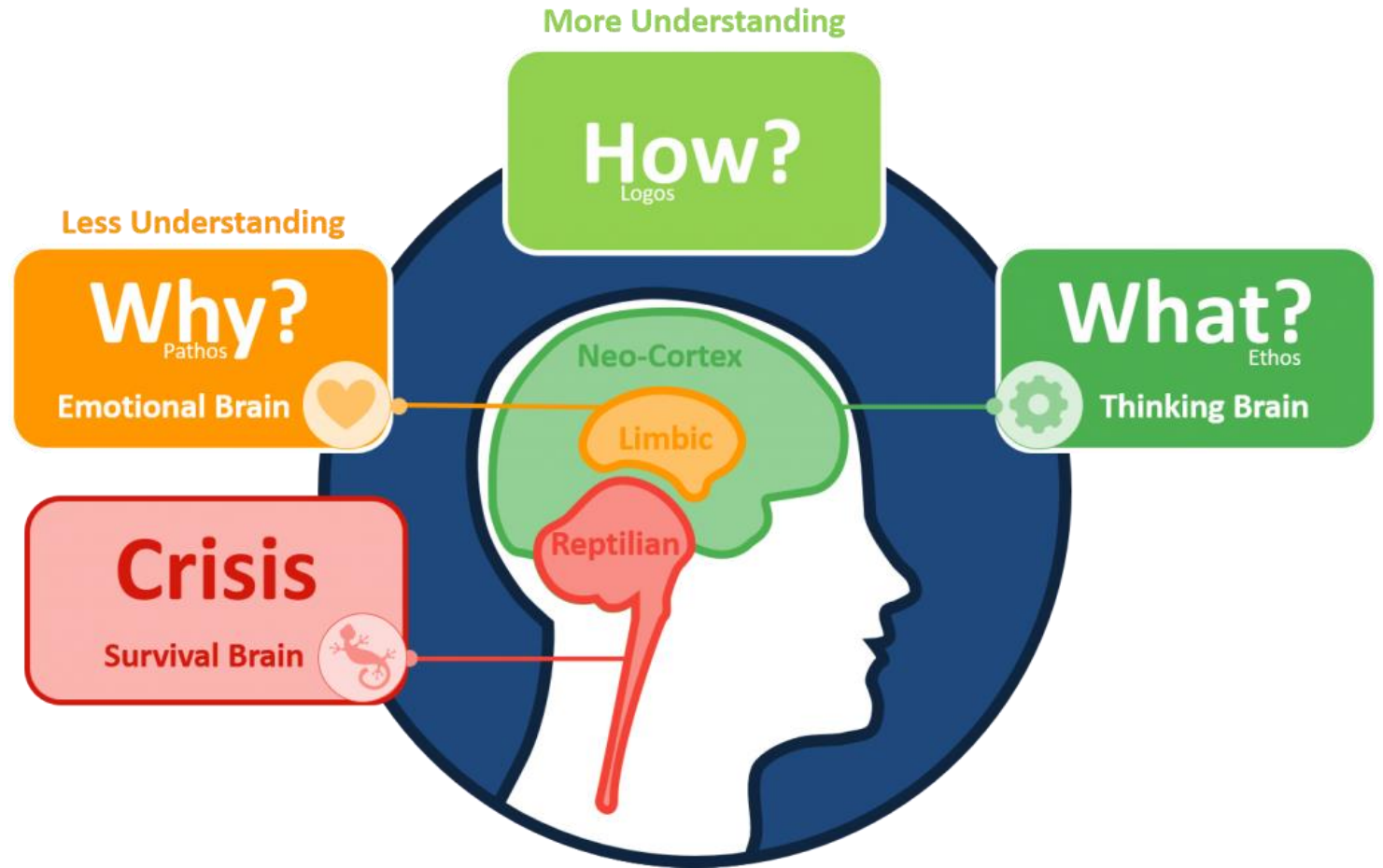
TRI-ONE BRAIN



SYSTEM 1/2

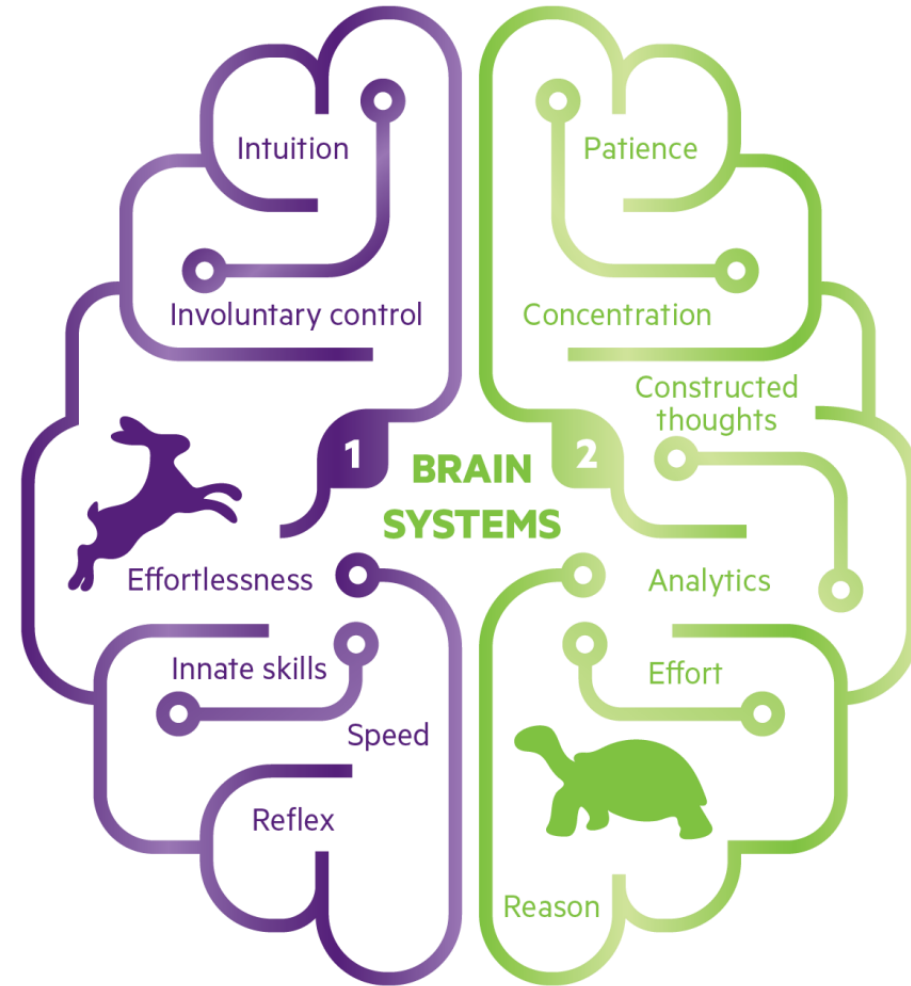
Triune Brain

(Thinking/(Re-)Acting)



With time habits (good/bad) becomes reptilian bias, without noticing you loose reasoning skills and ability to change at all.

System 1 and 2 (Thinking/(Re-)Acting)



Stimulation -> Time -> Reaction

*"People gather together with people
that make the work easier".*

Empathy

“Put yourself in the other shoes”

“Ask for reciprocity about it”



Resilience

Capacity to recover or thrive from any kind of difficulties in work and personal life





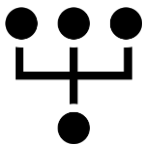
Change | Bi-Modality Awareness

Bimodal IT = Marathon Runners + Sprinters. Deeply Different, Both Essential

IT

Sys/Ops

Infrastructure



Maintenance,
Support

Mr. No

Think
Marathon
Runner



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Mode 1

Reliability

Price for
performance

Waterfall, V-Model,
high-ceremony IID

Plan-driven,
approval-based

Enterprise suppliers,
long-term deals

Good at
conventional
process, projects

IT-centric, removed
from customer

Long (months)

Goal

Value

Approach

Governance

Sourcing

Talent

Culture

Cycle Times

Mode 2

Agility

Revenue, brand,
customer experience

Agile, kanban,
low-ceremony IID

Empirical, continuous,
process-based

Small, new vendors,
short-term deals

Good at new and
uncertain projects

Business-centric,
close to customer

Short (days, weeks)

Think
Sprinter

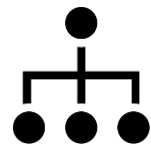


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Bussiness

Dev

Code



Innovation

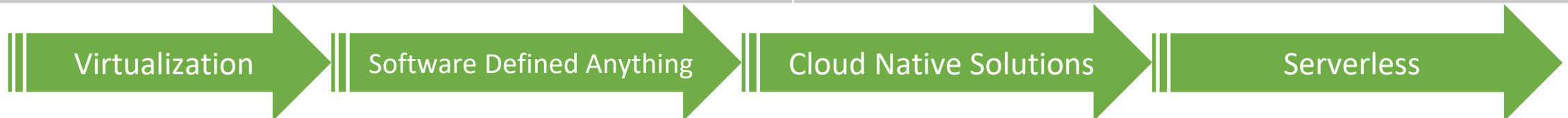
Mr. Yes

IaaS, Cloud, DevOps

Infrastructure vs. Code

Still slow, really slow,
transition...

Infrastructure	Code
Hardware / Software	Software
Static / Immutable / Un-versioned	Dynamic / Mutable / Versioned
Moore's Law (Faster)	Wirth's Law (Slower)
Extrinsic Documentation	Intrinsic Documentation
Administrator Driven (Indirect)	User Driven (Direct)





Changes | Way of Work



Changes

|

Way of Work

|

Organization

Technician vs. Engineer vs. Manager



Solves different problems



From technical to the business language



Doing, thinking or organizing people
(Hardly done in parallel by the same person)



From nothing to a book of knowledge

Managers (aka Coachs)

Deliver:

- Helps team members
- Drive the team to improve
- Support team good/bad times
- Organization, planification and auditing of the work progress

Expects:

- Proactive progress reporting
- Active participation in solutions
- Answers for follow-up question
- Ensure the job is done in time

Performance



Classification

Autonomy
Seniority
Delegation



Operation

Infrastructure
Code
Legacy



Documentation

Training
(Re-)usability
Auditability

Autonomy

(Freedom to innovate)



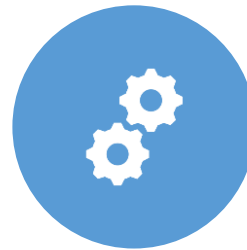
You work within a **team**
(not alone) for good & bad



You implement and
support **high-available**
solutions (whenever is
possible)



You foster **automation**
instead of manual task
(wherever is possible)



You look for **strategic approval**
(specially for new projects /
scope changes)

Seniority



Years doing same stuff → Changing on a daily basis



Just being different → Making the difference



Smart person → Smart team



A lot of Power → Responsibilities



Learning → Teaching *(New senior engineers will need it!)*

Delegation

Knowing what are your responsibilities is important

We are here because somebody let us make mistakes

Having time for innovation needs delegation and teaching to someone to be able to scale



Senior
Complex Tasks



Semi-Senior
Medium Tasks



Entry
Basic Tasks



2nd in Command (Backup)



RESPONSIBILITIES MATRIX



COUPLE OF YOUR SELECTION
(HIRE A NEW MEMBER?)

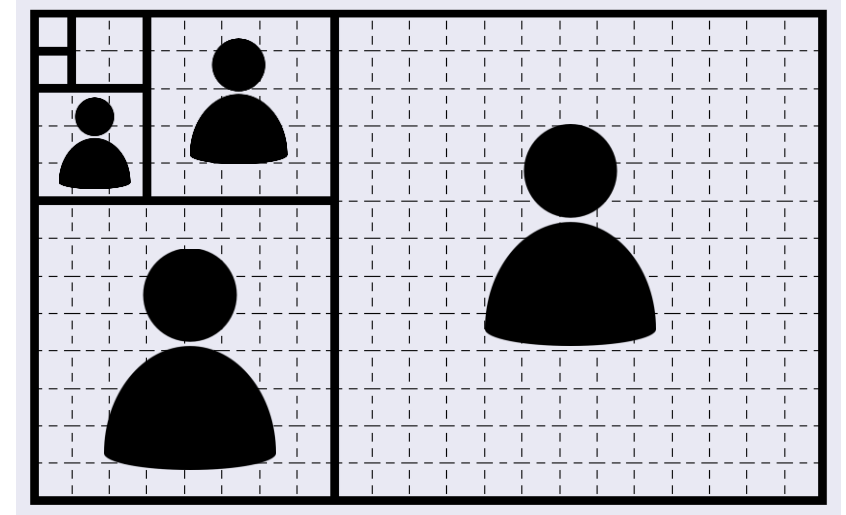


**PROMOTES SHARING OF
CRITICAL DOCUMENTATION
AND KNOWLEDGE**

Update Responsibility Matrix

Infrastructure:

- Network
- Compute
- Storage
- Management Systems
- Security/Identity Devices/Systems
- Backup System and Disaster Recovery
- Monitoring / Alerting (NPM, ITM, APM)
- Supporting Business Applications



Documentation (The Critical One)



Alive (always change)



Challenging for everyone (a lot of reasons)



Boring because it is not for you but for others



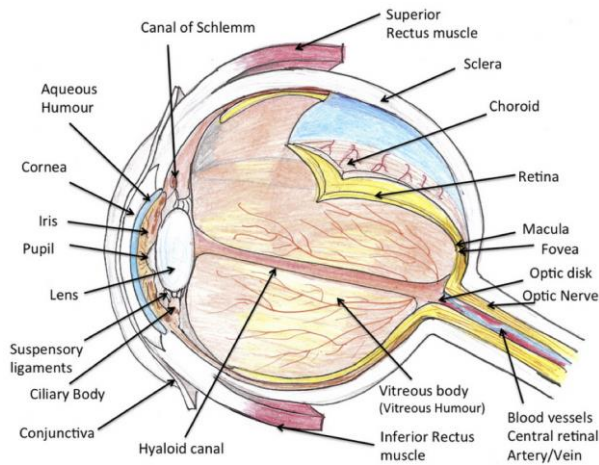
Needed for delegation to others (i.e. newcomers, leftovers when vacation, sickness, renounce).



Important when companies
growth up in time

Reduce tribal knowledge
Raise quality levels
Allow auditing

Documentation: Layered / Trimmed / Useful



Abstraction

KPI

Update -> Critical
Archive -> Obsolete

Focused on Training
Discoverable (Searchable)

Documentation (Workshops)



Resume: purpose, technology/vendor websites, external articles/references.



Architecture: high level (at least) visual representation of the platform or system (i.e. draw.io, dot/graphviz).



Assets: resources inventory labeling (not naming), links (URL) and credentials (e.g. Vault, SSO).



How-To's: about (re-)install and configure with focus on critical and tricky in-house customizations.



Basic administration and troubleshooting: standard procedures, know errors with brief solution explained, references to article/tickets and similars.



Changes

|

Way of Work

|

Innovation



Legacy

- There was, there is and there will be
- It's a matter of time, but it is important
- It is not neither bad nor good is just legacy
- Innovation is needed, but maintenance also

99% of the things are legacy since its go live

Innovation (& Investigation)



Must be focused

Value
(Customer)



Should be planned
(End/Start)

Trackable
(Timeframe)



Must be a process
(Success/Failure)

Measured
(Deliverable)



Changes

|

Way of Work

|

Communication

Communication

- **Notify:**

- Live changes with enough anticipation to customers
- Absence with anticipation (book and or block calendar)
- Delay/Leave to/from office to the team members
- If you are working and applying changes on weekend

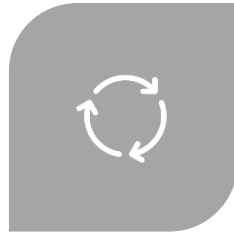
- **Avoid:**

- Doing unplanned changes not related to live issues on “Fridays”
- Implementing new features near days you are on vacations (freeze)
- Mixing informal/formal communication (e.g. me/x-team, misunderstanding, rumors)
- Overlapping vacations with your 2nd in command (*a.k.a. my backup*)

Estimated Time Ahead (ETA)



**BROADCAST
INFORMATION**
START, PROGRESS,
END, EVIDENCE



**REPORT
CONTINUOUSLY**
DON'T EXPECT TO
BE FOLLOW-UP



**RESPONSE
QUESTIONS**
TO EASY CLARIFY
FOLLOW-UP



**APPLY TO
EVERYTHING:**
INCIDENT,
PROBLEMS,
REQUIREMENTS.

Communication (Too many channels?)

@Email

- Many formal communications and calendaring
- Request to chats will be converted to tickets (as needed)

#Slack

- Live Issues and News
- Daily/Weekly Slack-Up Trial (instead of Daily Stand-Up)

JIRA (Tickets/Wiki)

- Kanban Board: Task/Requirements (Internal/External)
- Wiki: For publishing critical infrastructure information

Phone Calls / WhatsApp

- Only for emergencies

Mute Generation / Work Comm. Channel



**Favorite Comm.
Channel (IM)**

**Daily Comm.
Channel (IM)**

**Family Comm.
Channel (IM)**

So let's Slack-Up!



- (Trial) Broadcast to **slack channel** instead of verbal stand-up.
- **Daily/Weekly basis** at the start of the day before 10 A.M.
- **Reminder** for **achievements/contributions** to OKR's:
 - Maintenance
 - Innovation & Optimization
 - Support (**Only if not already a JIRA Ticket**)
 - Low, Lowest, Medium, High, Highest.
 - Undesired/Unplanned requests (e.g. *C-Level Urgencies, Last-Minute-Request*)
- Help to **track easily your progress and risks (red flags)**.
- Promote team **awareness of issues and progress**.

How it looks? <https://slack.com/intl/en-de/slack-tips/run-daily-standups-or-check-ins>



Changes

|

Way of Work

|

Planning

Planning

Objectives and
Key Results (OKRs)

In preparing for battle I
have always found that
plans are useless, but
planning is
indispensable.



Dwight D. Eisenhower
34th U.S. President
1890-1969

QuoteHD.com

Planning (OKR)



Maintenance Projects (30%)

Maintenance plans
Preventive monitoring



Optimization & Innovation Projects (20%)

Cost-efficiency driven plans
Security oriented driven plans
New technologies / features
try-out / deployment plans



Support Projects (50%)

New requirements
Reactive incident response

- Known errors handling
- Problems troubleshooting

Objectives & Key Results: Site Reliability Group

Mission: Ensure continuous operation and support for the core IT infrastructure at Westwing

On track
Watch out
Major issues

Objective Key Results

Progress

1. Maintenance Projects

Our continuous maintenance plans

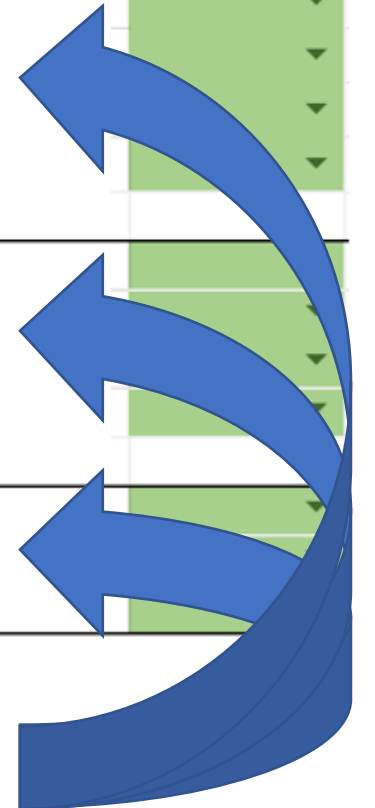
2. Optimization & Innovation Projects

Our innovation and optimization (micro-)projects

3. Support Projects

Refers to other tech teams support effort planned and unplanned but accounted

Initiatives: Pick up selection to improve operation?



Individual Planning / Report-Up

	URGENT	NOT URGENT
IMPORTANT	DO • Live Issue • Slack until fixed • Do the post-mortem documentation • OKR's • Slack-Up w/Manager	DECIDE • Non-Live Issue • Ask-for/Open ticket • Slack-Up w/Manager • Clarify expectations third parties (e.g. ETA, <i>Termin</i>) • Move to Initiatives
NOT IMPORTANT	DELEGATE • Teach fishing • Assign a less senior • 2nd in Cmd. (Backup)? • Manager	DELETE • Don't confuse it with Innovation at OKR's • Stop thinking on it

Estimated Time Ahead (ETA)

Never Ending Story

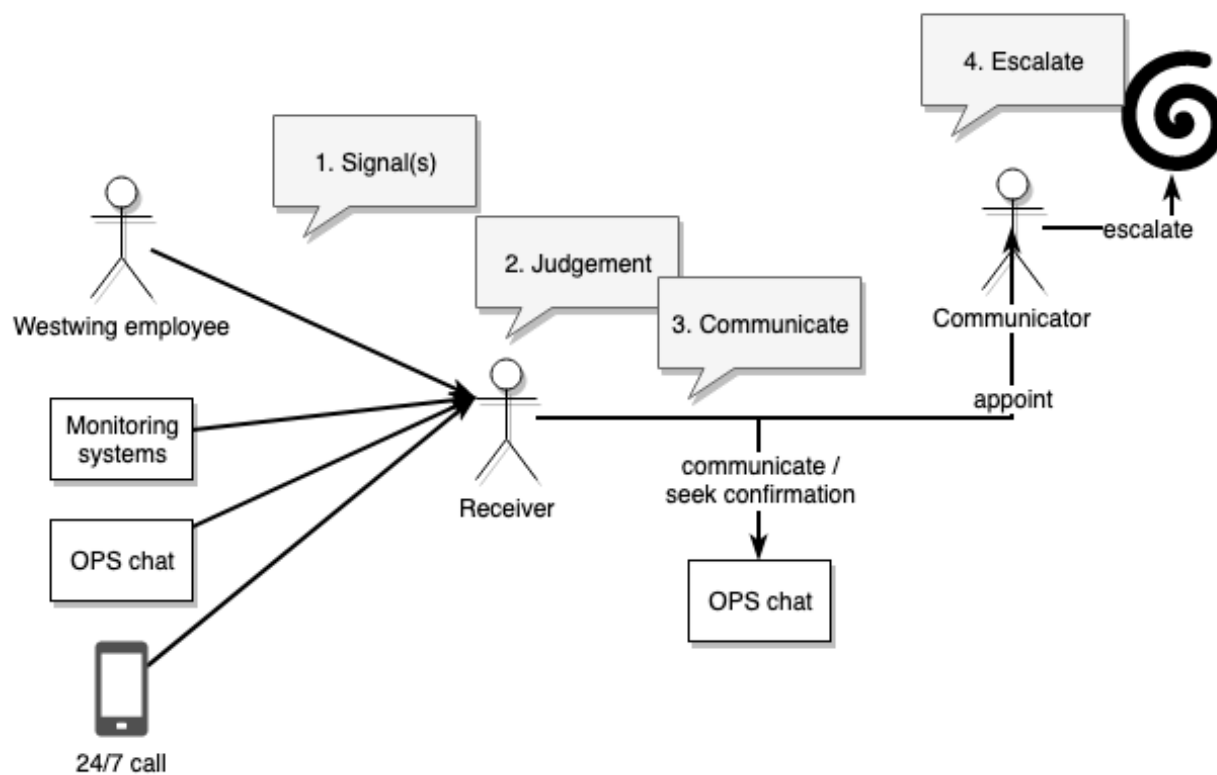
Unexpected
Additions/Drops

For critical
infrastructure is
important

Challenge for most
of the tech teams

Different for
innovation,
maintenance
and support

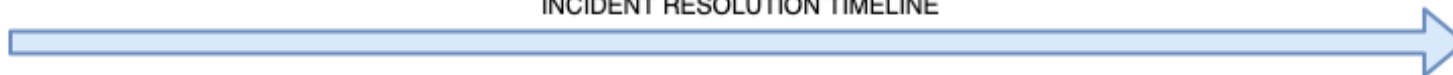
IT Incident Response Plan (Overview)



<45 minutes



INCIDENT RESOLUTION TIMELINE



Alerts: Person, Call, E-Mail, #Slack, SMS, without standard classification (severity).

Follow-up on slack Operational (OPS) channel (chat).

Direct Responsible Individual (DRI) take care of him/her alerts.

Meeting everyone involved in **Situational Room**.

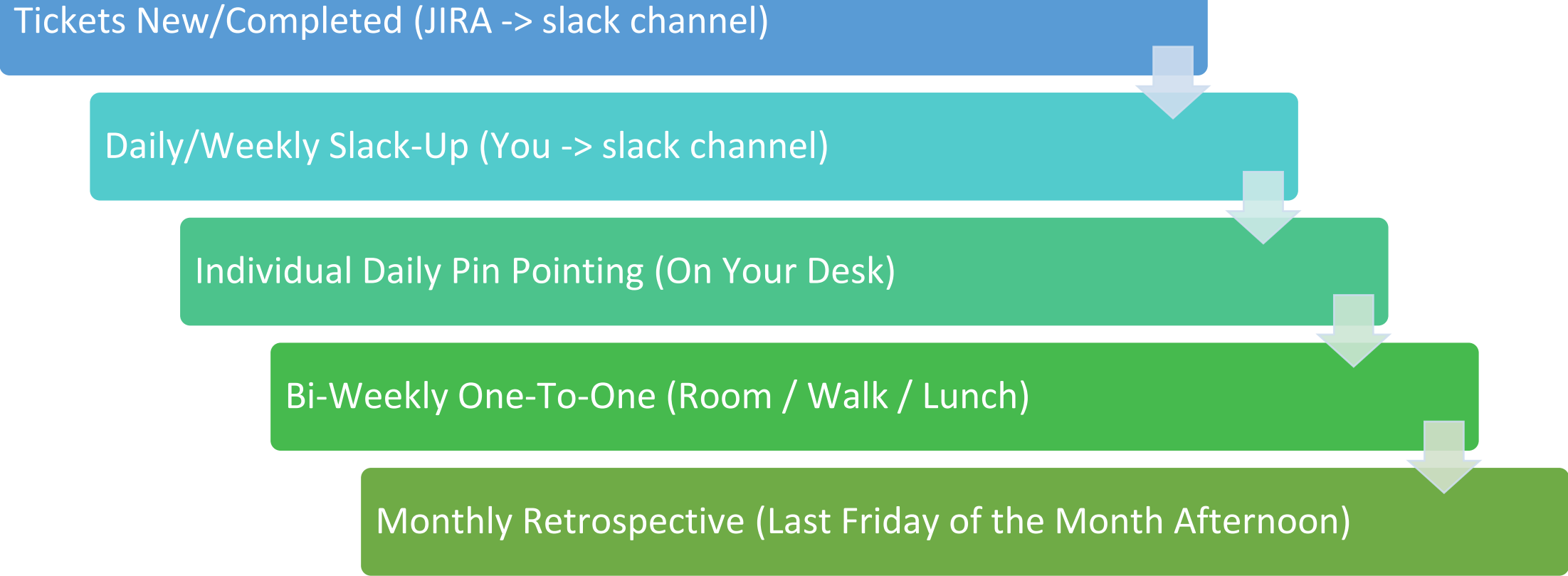
SLA defined in terms of maximum downtime in hours per system / application.

Spiral Escalation to Eng. Mgr., Head/Director, CTO, CEO.

IT Incident Response Insights



Tickets New/Completed (JIRA -> slack channel)



```
graph TD; A[Tickets New/Completed (JIRA -> slack channel)] --> B[Daily/Weekly Slack-Up (You -> slack channel)]; B --> C[Individual Daily Pin Pointing (On Your Desk)]; C --> D[Bi-Weekly One-To-One (Room / Walk / Lunch)]; D --> E[Monthly Retrospective (Last Friday of the Month Afternoon)];
```

Daily/Weekly Slack-Up (You -> slack channel)

Individual Daily Pin Pointing (On Your Desk)

Bi-Weekly One-To-One (Room / Walk / Lunch)

Monthly Retrospective (Last Friday of the Month Afternoon)

Progress (Follow-Up/Report-Up)



Trust &
Visibility

Trust & Visibility



Trust ring mitigates business hi-jacking



Critical credential and access levels should be shared (i.e. OneLogin) with key team members and C-Level (Breaking the glass)



Access should cover all infrastructure assets, platforms and systems



Monitoring tools and central consoles alerts for infrastructure must be broadcasted in communication channels (i.e. slack, e-mail)